Kexin Zhai . Jarvis Consulting

I studied at the University of Toronto (BSc, then MEng focused on analytics/ML), and I'm a data/analytics-oriented engineer with recent experience building small end-to-end systems and doing applied machine learning. On the technical side I work most comfortably in Python, Java, Linux/Bash, and SQL/RDBMS, and I've used Docker, PySpark, Databricks, and Azure for data workloads. I have built full-stack projects including a Linux cluster monitor, several quantitative and ML applications in Python, and database/web apps with designed schemas and wired up CRUD logic both academically and personally. My work experience backs that up: at GiftCash I worked with real marketplace data, pricing tools, SellerCloud, and repricing workflows; at the Ministry of Education I ran full QA cycles with Selenium, HP ALM, PL/SQL, and Azure DevOps to keep releases clean. I like staying in tech spaces that are practical, collaborative, and data-first. I care about showing I can execute, not just list tools.

Skills

Proficient: Python, PySpark, Databricks, SQL/RDBMS (PostgreSQL), R, Java, Git/GitHub, Apache Airflow, Apache NiFi, HashiCorp Vault, ETL/ELT Pipelines, Postman, REST API Integration, AI/NLP (LLMs, Multi-agent Architecture, OpenAI API, Prompt Engineering), Machine/Deep Learning (TensorFlow, PyTorch), Confluence, Lucidchart, Agile/Scrum, LaTeX, MATLAB, MS Office, G Suite

Competent: C, Starburst, Collibra, Data Modeling (SCD Type 2), Data & Cloud Analysis, Data Quality & Validation, Software QA & Automated Testing (pytest, Selenium, JUnit), Docker, Cloud Computing (AWS, Azure, GCP)

Familiar: Angular, Front-end Development (HTML/CSS, JavaScript, Bootstrap), Tableau, Power BI, Hadoop, MIPS, Linux/Bash, CI/CD

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_KexinZhai

Cluster Monitor [GitHub]: Built a lightweight Linux cluster monitoring solution that collects host hardware info and minute-level resource metrics with Bash scripts and ships them to a centralized PostgreSQL instance running in Docker. Automated data ingestion with cron, standardized the schema with SQL DDL, and provided sample analytical queries for capacity planning and troubleshooting. Tech stack: Bash, Docker, PostgreSQL, cron, Git/GitFlow, Linux command-line tools (lscpu, vmstat, df, awk).

Core Java Apps [GitHub]: Built a Java CLI grep application using Maven/IntelliJ, applying black-box testing to validate demo JAR behavior (regex matching, recursive file traversal, and output generation). Implemented RegexExc for pattern validation and designed JavaGrep/JavaGrepImp interfaces and classes to orchestrate file traversal, line filtering, and output writing with SLF4J/log4j logging. Enhanced the application using Java 8 Streams/Lambdas (LambdaStreamExc/Imp, JavaGrepLambdaImp), replacing loops with Files.walk/Files.lines. Packaged with Maven Shade, containerized using Docker (OpenJDK 8), tested with mounted data/log volumes, and published to Docker Hub. Tech stack: Java, Maven (Shade), IntelliJ, SLF4J/log4j, Java 8 Streams/Lambdas, Docker, Git/GitFlow.

Python Data Analytics [GitHub]: Built a retail analytics workflow in a Docker-based Python/Jupyter environment, connecting to PostgreSQL via psycopg2/SQLAlchemy and loading transaction data into Pandas. Performed SQL/Pandas data exploration and validation (pd.read_sql/pd.read_csv), calculated invoice totals, visualized amount distributions (full and 85th-percentile trimmed), and generated KPIs (orders, customers, SKUs, revenue, monthly sales). Extended analysis with time-series metrics (monthly orders/cancellations, sales growth, active/new/existing users) and RFM customer segmentation for retention insights. Tech stack: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, SQL, PostgreSQL, psycopg2, SQLAlchemy, Docker, Git.

PySpark Pipeline [GitHub]: Developed an end-to-end equity-market ETL pipeline ingesting daily OHLCV CSVs and company dimension data through a Raw → Bronze → Silver → Gold medallion architecture, producing analytics-ready datasets for performance/risk screening. Orchestrated Spark jobs with Airflow DAGs in Docker and implemented fail-fast data quality gates (schema/emptiness, ticker-date uniqueness, price/volume validity, trading-day completeness, sector/industry referential integrity), with audit/quarantine diagnostics for reproducibility. Generated Gold datasets with daily returns, rolling metrics, and sector/industry analytics (sector_daily_returns, industry_volatility). Tech stack: Python, PySpark, Apache Spark, Apache Airflow, Docker/Docker Compose, PostgreSQL, Parquet, exchange-calendars, SQL, pytest, Git.

Highlighted Projects

Adaptive Laddered ETF Momentum Strategy for Portfolio Optimization: Built a laddered momentum ETF strategy that used time-diversified tranches to enhance returns while reducing volatility and drawdowns; implemented a 2-tranche system with staggered 2-week investments to balance momentum capture and trend responsiveness; back-tested the strategy on sector ETFs (2020-2024) in Python and showed outperformance over SPY on risk-adjusted metrics (Sharpe/Sortino) with lower drawdowns; and automated weekly ETF selection plus reporting to demonstrate the laddering effectiveness for momentum portfolios.

Innovation Strategy Development for Canada: Guided an innovation-strategy study that compared Canada's innovation strengths (R&D, talent, institutions) with leading countries using Global Innovation Index data; collected and cleaned fragmented country-level innovation indicators to increase dataset reliability in Python (Pandas, Seaborn); applied clustering and feature-importance modeling to identify key drivers of innovation success across 100+ indicators; and developed an AI chatbot (Langflow + LLM) to deliver interactive insights and authored actionable policy recommendations.

Deepfake Detection with CNNs and Object Detection: Built a deepfake-detection pipeline using CNNs (InceptionV3, Xception, etc.) trained on 190K+ real/synthetic facial images; boosted accuracy by integrating YOLOv1 for face detection before classification; optimized performance through data augmentation and hyperparameter tuning while preventing overfitting; and tested model fairness across demographic groups and synthetic media types (GANs, StyleGAN).

Machine Learning Pipeline for Weekly Return Forecasting in Equity Markets: Built a Python-based weekly equity-return forecasting system that enhanced a textbook model with novel volatility/wavelet features; curated multi-asset data (Yahoo Finance/FRED) and engineered a boosted pipeline with Ridge/Elastic Net/Bayesian modeling; validated the models using Sharpe Ratio, Profit Factor, and anti-overfitting techniques (bootstrapping/Monte Carlo); and delivered a 40-slide presentation plus technical notebook that showed a production-ready ML workflow for finance.

APOPO Project on Rat Odor Preference Behavior for Landmine Detection: Partnered with APOPO to analyze rat-training video data and assessed how early exposure affects odor-preference behavior for landmine detection; processed large volumes of video-derived behavioral metrics to ensure reliability before statistical modeling; used PCA and mixed-effects models to identify key training factors and optimized model selection for interpretability; and delivered training recommendations and co-authored a report to enhance APOPO's conditioning program.

Bullet Journal Webpage Application [GitHub]: Built a bullet-journal web app using Angular and Agile/Scrum, finishing frontend tasks from the backlog so the UI stood out from other diary apps; conducted user-centric testing to keep the features aligned with stakeholder standards and reported issues to the team for timely fixes; and tracked project progress with CI/CD tools, shared workflows, and collaboration with frontend/QA teams via Zoom, GitHub, and Discord.

Professional Experiences

Data Engineer, RBC Wealth Management (Royal Bank of Canada) (January 2026 - October 2026): Collaborated with cross-functional teams to maintain dozens of daily ETL pipelines, troubleshooting production issues, resolving bugs, and performing upgrades to ensure operational stability; designed and developed end-to-end ETL pipelines using Python, SQL, Airflow, NiFi, and PySpark, translating business requirements into technical workflows and presenting designs; built metadata ingestion pipelines integrating Collibra REST APIs with PostgreSQL, implementing data transformations, SCD Type 2 modeling, and incremental data processing for historical tracking; performed data quality checks, schema validation, and troubleshooting to ensure data accuracy, consistency, and pipeline reliability; leveraged GitHub, Confluence, Lucidchart, Starburst, and Vault while following enterprise SDLC, testing, and change management practices across environments.

Junior Pricing Analyst, GiftCash (September 2023 - December 2023): Leveraged discounted gift cards and product-research tools such as JungleScout and Keepa to identify high-ROI Amazon products, and analyzed Buybox competition to determine optimal FBA/FBM strategies using a custom ROI calculator; managed product data in SellerCloud and coordinated online/offline purchasing teams to keep inventory at optimal levels while processing real-time buying decisions based on profitability analysis; monitored and repriced Amazon listings through Repricer, managed eBay listings with cross-platform price synchronization, and liaised with warehouse teams to ensure fulfillment compliance across both marketplaces.

IT Quality Assurance Assistant, Ministry of Education (September 2021 - April 2022): Performed manual and automated testing for Ministry of Education applications using Selenium, HP ALM, and PL/SQL Developer, ensuring defect-free production releases while managing multiple projects and maintaining communication with cross-functional teams; executed the full testing lifecycle including requirements review, test planning with traceability matrices, test-case execution in HP ALM, and results analysis, while using Azure DevOps to log defects and follow up with development

teams for resolution; mastered HP ALM, PL/SQL Developer, and Selenium frameworks to optimize testing efficiency, implemented automated regression testing, and contributed to process improvements through DevOps collaboration and documentation standardization.

Education

University of Toronto (2024-2025), Master of Engineering, Mechanical & Industrial Engineering - Technical Emphasis in Data Analytics and Machine Learning, MEng Certificate in Financial Engineering - GPA: 3.97/4.0

University of Toronto Scarborough (2019-2024), Honours Bachelor of Science (Coop) with High Distinction, Computer and Mathematical Sciences - Specialist in Statistics (Machine Learning & Data Mining) with Major in Computer Science & Minor in Linguistics - Dean's List (2019, 2020, 2021, 2022, 2023, 2024) - GPA: 3.83/4.0

Miscellaneous

- Quantium Data Analytics Virtual Experience Program on Forage
- KPMG Data Analytics Consulting Virtual Internship on Forage
- BCG Data Science & Analytics Virtual Experience Program on Forage
- IBM Business Analyst Work Placement Education Program: Digital Transformation – The Road to Future-Ready Organizations
- Vice President, Chinese Students and Scholars Association at University of Toronto Scarborough: Negotiated sponsorships, drafted contracts, led directors in task delegation, and fostered sponsor relations to boost mutual recognition.
- Volunteer Leader, Math in Motion - Girls in Gear!: Guided the new student leaders to take action within groups harmoniously; coached leaders to learn new design tools.